# Flow

## 🧠 UPDATED SYSTEM ARCHITECTURE (FINAL VERSION)

```
[ Mobile App ]
   ↓
[ Backend Server ]
   ↓
[ Database ]
   ↓
[ Notification Engine ]
   ↓
[ Nearby Users (Community Network) ]
```

❗ Removed:

- ❌ Voice trigger for activation
- ❌ Separate volunteer system
- ❌ Admin in emergency flow

👉 Now it's **faster, simpler, and more scalable**

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## 📱 1. MOBILE APP ARCHITECTURE (UPDATED)

### 🔁 1. Background Service (Always Running)

This is your **core engine**

It listens ONLY for:

- Power button (pressed 3 times)

- Custom shortcut (user-defined combination)

👉 Example:

- Power + Volume Down
- Volume Up 3 times

⚠️ No voice trigger here (as you decided)

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## ⚡ 2. Trigger Detection Module (UPDATED)

```
IF user-defined shortcut detected
→ Activate Emergency Mode
```

👉 Important Feature:

- User can **customize trigger**
- Makes app **flexible + user-friendly**

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## 🧠 3. Validation Engine (Same Concept, Cleaner Flow)

After activation:

### Step-by-step:

```
Step 1 → Quick popup (tap / pattern)

IF success → Emergency Confirmed

ELSE

Step 2 → Shake detection

IF success → Emergency Confirmed

ELSE
```

```
Step 3 → Voice keyword detection (ONLY AFTER ACTIVATION)

IF success → Emergency Confirmed

ELSE

Step 4 → AI fallback
```

👉 Voice is now **validation-only**, not trigger

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## 🎥 4. AI Capture Module

If user cannot respond:

- Record:
  - 🎤 Audio (5 sec)
  - 📷 Video (5 sec)

Send to backend

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## 📍 5. Location Module

- Fetch current GPS
- Start live tracking

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## 🚨 6. Emergency Sender

Send:

```
{
  user_id,
  live_location,
  timestamp,
  verification_type,
```

```
  media (optional)
}
```

# 🌐 2. WEBSITE ARCHITECTURE (UPDATED & SIMPLIFIED)

## 🎯 Purpose:

ONLY for:

- User setup
- Monitoring reports (Admin)

## Modules:

### 👤 1. Unified User System (IMPORTANT CHANGE)

- No separate volunteer role
- Every user = potential helper

Each user can:

- Send emergency
- Receive nearby alerts

👉 This is **peer-to-peer safety network**

### 📋 2. Emergency Contacts Setup

- Add family/friends
- Manage contacts

### 🛠️ 3. Admin Dashboard (LIMITED ROLE)

Admin can:

- View reports
- Track system activity

Admin CANNOT:

- Interfere in real-time emergency
- Be part of alert chain

👉 Smart decision → reduces dependency

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# ⚙️ 3. BACKEND ARCHITECTURE (UPDATED LOGIC)

## 🚨 1. Emergency Handler (CORE)

When alert comes:

```
1. Save emergency
2. Fetch nearby users
3. Send alerts directly
```

👉 No admin in between

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## 📍 2. Nearby User Finder

- Finds users within radius (e.g., 2–3 km)

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## 📡 3. Notification System

Sends:

- 📲 Push notifications
- 📩 SMS (optional)

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## 🤖 4. AI Service (Fallback Only)

Used ONLY if:

- User fails validation

Output:

```
{
  is_emergency: true/false,
  confidence
}
```

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## 🔄 5. Live Tracking Service

- Updates location continuously
- Shared with nearby users

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# 🗄️ 4. DATABASE (UPDATED)

### Users

```
user_id
name
phone
current_location
```

---

### Emergency Contacts

```
user_id
contact_phone
```

---

### Emergencies

```
emergency_id
user_id
location
status
verification_type
timestamp
```

# 🔁 5. FINAL WORKFLOW (UPDATED — VERY IMPORTANT)

## 🚨 Scenario: User in danger

### STEP 1: Trigger

User presses:

👉 Power button 3 times (or custom shortcut)

### STEP 2: App Activation

- Background service activates app
- Emergency mode ON

### STEP 3: Validation Flow

```
Popup → FAIL
Shake → FAIL
Voice → FAIL
```

👉 AI activated

### STEP 4: AI Analysis

- Audio/video recorded
- Sent to backend
- AI confirms emergency

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## STEP 5: Emergency Created

Backend logs event

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## STEP 6: Nearby Users Identified

- Within 2–3 km radius

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## STEP 7: Alerts Sent (DIRECT — NO ADMIN)

Each nearby user gets:

- 📍 Live location
- 🚨 Emergency alert
- 🎥 Optional media

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## STEP 8: Live Tracking

- Continuous location updates
- Shared with responders

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## STEP 9: Response

Nearby users:

- Can reach location
- Can contact victim

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## STEP 10: End

- User marks safe
  OR

- System auto closes

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# 🔥 WHY THIS VERSION IS STRONGER (JUDGE POINTS)

Your improvements made it:

## ✅ More Practical

- No unreliable voice trigger

## ✅ Faster Response

- Direct P2P alerts (no admin delay)

## ✅ Scalable

- Works like a network (like Uber model)

## ✅ Simple Architecture

- Easy to build in hackathon

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# 🎯 HOW TO EXPLAIN THIS IN 1 LINE

👉

**“We built a peer-to-peer emergency network where any user can instantly trigger help using customizable shortcuts, validated through multi-layer checks, and directly alert nearby users in real-time without any dependency on authorities.”**

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